

Web Programming(CS4032)

Sessional-2 Exam

Date: April 9, 2025

Total Time: 1 hour

Course Instructor(s)

Total Marks: 25

Mahhek Tahir, Saad Farooq

Total Questions: 3

Roll No

Section

Student Signature

- Attempt all questions on the answer sheet
- ONE A4 Size **handwritten** cheat sheet with **Roll#** written on top of the sheet is allowed.

CLO 3: Analyze user requirements and web development problems to design solutions

Question 1

[3 marks]

Identify the correct React Hook. Provide a small code snippet demonstrating its usage

1. You want to fetch data from an API when the component mounts.
2. You need to store a reference to a DOM element so you can focus an input field when the page loads.
3. You need to store a user's input in a form field and update the UI as they type.

CLO 3: Analyze user requirements and web development problems to design solutions

Question 2

[2 marks]

Review the following JSX code to identify the error.

```
const Example = () => {  
  return (  
    <h1>Hello</h1>  
    <p>Welcome to React!</p>  
  );  
};  
export default Example;
```

CLO 4: Develop dynamic, server-side or client-side applications using modern web frameworks and databases

Question 3

[20 marks]

You need to build a Product Cart System using *React*, *Redux Toolkit*, and *React Router*.

Given Data (product.js)

You are provided with a **product.js** file that contains the following product data:

```
export const products = [  
  { id: 1, name: "Laptop", price: 1200 },  
  { id: 2, name: "Phone", price: 800 },  
  { id: 3, name: "Headphone", price: 200 },];
```

National University of Computer and Emerging Sciences

Tasks:

1. **Product Listing Page (Home Page) (5 marks)**
 - Display a list of products (name, price) from **product.js**. (Remember to import the file)
 - Each product should have an **"Add to Cart"** button.
 - Do not worry about multiple clicks to the same product.
2. **Cart Page (5 marks)**
 - Display a list of products that were added to the cart.
 - Each cart item should have a **"Remove from Cart"** button to remove it from the cart.
3. **Redux Setup (5 marks)**
 - Implement **cartSlice.js** using **Redux Toolkit** with the following actions:
 - **addToCart(product)**: Adds the selected product to the cart.
 - **removeFromCart(id)**: Removes the product from the cart.
4. **Routing (App.js) (5 marks)**
 - Use React Router to create routes for:
 - **Home Page ("/")** → Displays the product listing.
 - **Cart Page ("/cart")** → Displays the cart items.
 -
 - Configure the Redux store inside **App.js** and wrap the app with **<Provider>**.

```
import { BrowserRouter as Router, Routes, Route, Link } from
"react-router-dom";
import { Provider } from "react-redux";
import { configureStore } from "@reduxjs/toolkit";
import cartReducer from "./cartSlice";
import ProductList from "./ProductList";
import CartPage from "./CartPage";

/* Configure Redux store here */
const App = () => {
  return (
    <>
      { /* Wrap with Provider and Redux store */ }
      { /* Setup Router */ }
      { /* Navigation Links */ }
      { /* Define Routes */ }
    </>
  );
};
```

Requirements:

- Use **React Hooks** (e.g., **useState**) where applicable.
- Use **Redux Toolkit** for state management (**cartSlice.js**).
- **Do not create a separate store.js** – configure the store inside **App.js**.
- Use **React Router** for navigation.
- **No CSS or SCSS files** (Styling is not required).